The Price of Fairness, Complexity, and Incentives in No-Numeraire Combinatorial Trade-Intent Auctions

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Abstract. Blockchain trade-intent auctions, which currently intermediate on the order of ten billion dollars in trades per month, are combinatorial because executing several intents jointly produces extra efficiencies, yet they have *no numeraire*: there is no common unit in which to redistribute those efficiencies. The deployed fix builds an *endogenous fairness benchmark* from the bids and admits a batched bid only if it beats that benchmark for every covered trader. We give a worst-case theory of this design around one invariant: the complementarity factor g, the largest ratio of a batch's value to the sum of its components' standalone values. The price of fairness is exactly $\min(g, 1/\theta)$ in liquidity $\theta = 1 - \lambda$, tight for all parameters and independent of the number of orders and of prices; the endpoints give a dichotomy between g (no numeraire) and 1 (full numeraire). Batch size is irrelevant; the right refinement is a per-leg coverage floor α, under which the exact law is $\min(g, \alpha + (1-\alpha)/\theta)$. A Walrasian/UDP equilibrium supporting an efficient allocation exists for every instance *iff* $g = 1$, by native no-numeraire arguments, and at the trader-optimal equilibrium the price system *is* the fairness benchmark. The price of strategyproofness is $\Theta(1 + \log g)$ for a single multi-minded solver and for many single-minded solvers, via two mechanisms and a convex profit-curve technique; the remaining multi-agent multi-minded cell admits $O(N \log g)$ and a matching $\Omega(\min(N, g))$ barrier for serial-demand mechanisms. Online, the price splits: against realized lifetime floors even randomization pays exactly g, while with static public floors the residual problem is one-way search over the spread, deterministic $\Theta(\sqrt{g})$ and randomized $\Theta(\log g)$. Finally, manipulating the benchmark itself is a priced game with a closed-form bias and a sharp participation threshold. Throughout, one scale-free family governs the informational and the incentive price of the spread g.

Keywords: Combinatorial auctions · Price of fairness · Mechanism design · Blockchain · Competitive analysis.

1 Introduction

Blockchain "trade-intent" auctions let users state a desired exchange of one asset for another and delegate execution to competing agents called *solvers*; the protocol collects intents off-chain and auctions them in batches. These markets are economically significant— recent figures put monthly volume on the order of ten billion dollars[1]—and they are *combinatorial*: a solver executing several intents together can net opposing flows and save on fees, producing more value than executing them separately. What makes them unlike the combinatorial auctions of the literature is the absence of a *numeraire*. Users may demand any asset for any other, there is no common unit of account, and assets are illiquid, so the extra value created by batching cannot simply be paid out and shared.

The deployed response (the *fair combinatorial auction*, adopted by the studied protocol) constructs a fairness benchmark *from the bids themselves*: each trader's standalone competitive outcome defines a floor, and a batched bid is executed only if it delivers at least that floor to *every* covered trader. This sidesteps the impossibility of sharing batching gains by never allowing a batch to make any trader worse off than individual competition would. The equilibrium behavior of this design was analyzed by Canidio and Henneke [5], who identify a tension between the fairness guarantee and the value returned to traders. What is missing is a *worst-case* theory: how much welfare does the fairness filter cost, when do competitive equilibria exist, what does strategyproofness cost, and how do these prices behave online and under manipulation.

One invariant. We organize such a theory around a single structural quantity. For a solver with batch value $V_i(S)$ on a set S of intents and standalone values $V_i(\{j\})$, the *complementarity factor* is the least $g \geq 1$ with $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ for all i, S: how much more a batch can be worth than the sum of its parts. The benchmark floor of order j is $c_j = \max_i V_i(\{j\})$, and liquidity is $\theta = 1 - \lambda \in [0, 1]$, where λ is the haircut on moving value across assets ($\theta = 0$ is the pure no-numeraire regime, $\theta = 1$ restores a numeraire). We show that g—together with θ—controls a remarkably wide range of worst-case quantities, which is what gives the design a unified analysis rather than a collection of unrelated bounds.

Contributions.

1. **Price of fairness (Section 3).** The welfare ratio between the unconstrained optimum and the best fair outcome is exactly $\mathrm{PoF}(\theta, g) = \min(g, 1/\theta)$, tight for all θ, g and independent of the number of orders and of prices, with a no-numeraire/ numeraire dichotomy $\{g, 1\}$. The binding instance is an asymmetry at the *repair/drop indifference point*, not at maximal complementarity. Batch size is not the right parameter; under a per-leg coverage floor α the exact law is $\min\big(g, \alpha + \frac{1-\alpha}{\theta}\big)$, pricing the worst single-leg shortfall discounted by liquidity.
2. **Equilibrium existence and the core (Section 4).** A Walrasian/UDP equilibrium supporting an efficient allocation exists for every instance *iff* $g = 1$, both directions proved natively in the no-numeraire model; at the

[1] Order-of-magnitude figure from the deployment literature on the studied protocol; the precise figure and asset counts are recalled in the full version.

Table 1. Worst-case prices as functions of the complementarity factor g and liquidity θ (N: number of strategic solvers; α: coverage). "exact" marks tight characterizations.

Quantity	Result	Status
Price of fairness	$\min(g, 1/\theta)$; endpoints $\{g, 1\}$	exact
coverage refinement	$\min(g, \alpha + (1-\alpha)/\theta)$; size-irrelevant	exact
Walrasian/UDP existence	for all instances iff $g = 1$	exact
Core	filter = singleton-core; g-core nonempty, tight	exact
Winner determination	NP-hard; FPT in directed pairs P	—
Strategyproofness (core)	$\Theta(1 + \log g)$; det. exactly $\sqrt{g}$	exact
single-/multi-minded	both $\Theta(1 + \log g)$; indep. $\Omega(g/\log^2 g)$	exact
multi-agent multi-minded	$O(N \log g)$ / serial $\Omega(\min(N, g))$	partial
Online (realized floors)	exactly g, even randomized	exact
Online (static floors)	det. $\Theta(\sqrt{g})$, rand. $\Theta(\log g)$	exact
Price of anarchy	individual-only $\leq e/(e-1)$; full conj. $\Theta(g)$	partial
Contamination	threshold $\kappa_1 \geq v\beta$ deters; $g \mapsto g/(1-\mu)$	—

trader-optimal equilibrium the obligations are exactly the fairness floors. The deployed filter is precisely the singleton-coalition core constraint, the individual allocation always lies in the multiplicative g-core (tight), and the no-numeraire core can be nonempty where the transferable core is empty.

3. **Price of strategyproofness (Section 5).** In an exogenous-benchmark regime that the impossibility results naturally motivate, the price of strategyproofness is $\Theta(1 + \log g)$ both for many single-minded solvers and for one multi-minded solver, via an eligibility-posted packing mechanism and a coupled-menu mechanism with a convex profit-curve argument; independent option pricing instead pays $\Omega(g/\log^2 g)$. The remaining multi-agent multi-minded cell is pinned between $O(N \log g)$ and an $\Omega(\min(N, g))$ barrier for serial-demand mechanisms.
4. **Online and strategic prices (Section 6).** Deterministic online fair settlement has competitive ratio exactly g; randomization does not help against realized lifetime floors, but once floors are static and public the residual problem is one-way search with deterministic price $\Theta(\sqrt{g})$ and randomized price $\Theta(\log g)$. The first-price game's individual restriction is smooth (coarse-correlated price of anarchy $\leq e/(e-1)$); a contamination game prices manipulation of the benchmark, with a sharp participation threshold and a rate-limit "ramp tax."

We also locate the complexity of fair winner determination: it is NP-hard but fixed-parameter tractable in the number of directed asset pairs. Several results expose a recurring phenomenon—the no-numeraire constraint *removes deviations*—and a single geometric hard family that governs both informational and incentive prices. Table 1 collects the quantitative results.

Related work. The model and the equilibrium analysis of the fair combinatorial auction are due to Canidio and Henneke [5]; we complement their 2×2 equilibrium study with a worst-case, approximation-theoretic treatment. Combinatorial auctions are classical [7], but almost always *with* a numeraire and quasilinear utilities; our setting has neither. The *price of fairness* was introduced for divisible resources [3] and developed for indivisible goods under proportionality, envy-freeness, and MMS [6,2,1], where it scales with the number of agents; our price of fairness is a different object—it scales with production complementarity and liquidity, not agent count, because the efficiency loss comes from the absence of a numeraire rather than from indivisibility. Equilibrium existence draws on assignment markets [9] and the package-assignment characterization [4], but neither applies verbatim without transfers, and we give native arguments. Our smoothness bound uses the first-price deviation of Syrgkanis and Tardos [10]; the online search bounds instantiate one-way trading with a known spread [8].

2 Model

There are n trade intents (*orders*) and a set of *solvers*. A solver i submits bids: an *individual* bid on a single order, or a *batched* bid (i, S, y) specifying a set S of orders and, for each $j \in S$, a delivered value v_j in j's requested asset. The solver's value for the batch is $V_i(S) = \sum_{j \in S} v_j$ when it executes; production is asset-specific, so absent conversion a winner delivers $w_j \leq v_j$ to each $j \in S$ *in asset* a_j and cannot move value across assets or traders. There are no monetary transfers— the structural element absent from classical combinatorial auctions.

Definition 1 (Complementarity factor). *The instance has complementarity factor* $g \geq 1$ *if* $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ *for every solver* i *and every* S*. Thus* g *caps how much batching amplifies standalone value;* $g = 1$ *is additive.*

Definition 2 (Fairness floor; fair allocation). *The floor of order* j *is* $c_j := \max_i V_i(\{j\})$*, the value of its best standalone bid;* $C := \sum_j c_j$*. An allocation is fair if every allocated order receives* $w_j \geq c_j$*; batched bids violating this for some covered order are filtered.*

An allocation selects winning bids covering disjoint order sets; its welfare is $W = \sum_j w_j$. Let OPT be the maximum welfare ignoring fairness and FAIR the maximum over fair allocations; the *price of fairness* is $\mathrm{PoF} := \sup \mathrm{OPT}/\mathrm{FAIR}$ over a stated class.

Definition 3 (Liquidity). *Within a single winning bid a solver may convert value across its traders at haircut* $\lambda \in [0, 1]$*: withdrawing* x *from one trader's asset yields* $(1-\lambda)x$ *for another. Write* $\theta := 1 - \lambda$ *for liquidity;* $\theta = 0$ *is the pure no-numeraire case,* $\theta = 1$ *a full numeraire. Conversion is per-bid.*

The deployed mechanism additionally requires uniform clearing prices per directed token pair (*uniform directed prices*, UDP), forcing all orders on one directed pair into a single winning bid. UDP is vacuous in Sections 3–5 (our constructions use distinct pairs) and reappears as the source of fixed-parameter tractability. Full proofs of all numbered results are in the appendix.

3 The Exact Price of Fairness

Theorem 1 (Price of fairness, exact). *For every $\theta \in [0,1]$ and $g \geq 1$, over all instances with liquidity θ and complementarity factor at most g,*

$$\mathrm{PoF}(\theta, g) = \min\bigl(g,\, 1/\theta\bigr),$$

tight, and independent of the number of orders and of prices ($1/\theta := \infty$ at $\theta = 0$).

Proof (Proof idea). Upper bound. Process each bid of an unconstrained optimum independently. A bid with gain $\gamma = V/C_S \in [1, g]$ has deficit $D = \sum_{j \in S}(c_j - v_j)^+$; *repairing* it by withdrawing D/θ from its surplus to top up starved legs is feasible when γ is large, yielding welfare $\geq V/\min(g, 1/\theta)$, and otherwise *dropping* to floors gives $C_S = V/\gamma \geq V/\min(g, 1/\theta)$. Summing over the disjoint optimal bids gives $\mathrm{FAIR} \geq \mathrm{OPT}/\min(g, 1/\theta)$. *Lower bound.* A two-order instance with one starved leg, calibrated to the repair/drop indifference point $\gamma^* = \min(g, 1/\theta)$, forces $\mathrm{FAIR} = c$ against $\mathrm{OPT} = \gamma^* c$. The binding instance is an extreme asymmetry, not maximal complementarity. ☐

Corollary 1 (Dichotomy). *At the endpoints $\mathrm{PoF} \in \{g, 1\}$: the pure no-numeraire auction ($\theta = 0$) pays the full factor g, while any numeraire ($\theta = 1$) removes the fairness cost entirely.*

A natural attempt to add a third parameter fails, and identifies the right one.

Proposition 1 (Batch size is not the parameter). *With $\mathrm{PoF}(\theta, g, d)$ the price restricted to bids covering at most d orders, $\mathrm{PoF}(\theta, g, 1) = 1$ and $\mathrm{PoF}(\theta, g, d) = \min(g, 1/\theta)$ for every $d \geq 2$.*

Say an instance has *coverage* $\alpha \in [0,1]$ if every bid satisfies $v_j \geq \alpha c_j$ on every covered order—no leg is worse than α times the benchmark, a per-leg routing-quality floor.

Theorem 2 (The coverage law). *Over instances with liquidity θ, complementarity at most g, and coverage α,*

$$\mathrm{PoF}(\theta, g, \alpha) = \min\Bigl(g,\ \alpha + \tfrac{1-\alpha}{\theta}\Bigr),$$

with $\Gamma(0, \alpha) = \infty$ for $\alpha < 1$ and $\Gamma(0, 1) = 1$; tight on two-order instances.

The quantity $\Gamma - 1 = (1 - \alpha) \cdot \frac{1-\theta}{\theta}$ is the largest relative per-leg shortfall times the conversion loss per unit of delivered deficit: the price of fairness prices the worst starved leg, discounted by liquidity. The endpoint $\alpha = 1$ is exactly the componentwise filter-passing under which fairness is free.

4 Equilibrium Existence and the Core

We work in the order-price abstraction of UDP: a directed-rate system induces order-level obligations $w_j \geq 0$ that any solver serving j must deliver, and the single property used is that w_j depends only on the order. A *UDP equilibrium supporting the efficient allocation* is a rate system and an allocation in which every order is served at exactly its w_j, every solver's executed portfolio maximizes its profit $\sum_j (v_j - w_j)$ over feasible (executable, i.e. $v_j \geq w_j$) portfolios, and produced value equals OPT.

Lemma 1 (Batch domination under $g = 1$). *If $g = 1$ then at any rates every executable batched bid is weakly dominated by the sub-portfolio of its positive-profit singletons: $\sum_{j \in S}(v_j - w_j) = V_i(S) - \sum_j w_j \leq \sum_j (V_i(\{j\}) - w_j)^+$. Hence every solver has an individual-only best response; the one use of UDP is that w_j is the same in a batch or a singleton.*

Theorem 3 (Existence frontier is $g = 1$). *A UDP equilibrium supporting an efficient allocation exists for every instance of the class with complementarity at most g iff $g = 1$. For $g = 1$ the obligations $w_j = c_j$ are an equilibrium, so the equilibrium is best-bid fair; for every $g > 1$ some instance admits none. (Solvers have no capacity constraints; in (a) the floor vector is assumed rate-realizable.)*

Proof (Proof idea). *$g = 1$:* set $w_j = c_j$ and assign each order to a best producer; winners earn zero margin, losers nonpositive, and Lemma 1 kills batch deviations. Produced value is $C = \text{OPT}$. *$g > 1$:* natively—non-existence with transfers does not imply it here, since feasibility only *removes* deviations—a three-cycle of overlapping pair-batches with vector $(1 + \frac{\delta}{2}, 1 + \frac{\delta}{2})$ over floors 1: individual feasibility forces $w_3 \leq 1$, which blocks the high-rate support route, and the remaining deviation inequalities give $\delta \leq 0$. □

The filter also has a cooperative reading. A bid *blocks* an allocation at factor κ if some feasible delivery gives every covered order strictly more than κw_j; the κ-core forbids this.

Proposition 2 (Core placement). *(i) Best-bid fairness is exactly immunity to singleton blocking at factor 1: the deployed filter is the singleton-coalition core constraint. (ii) The individual allocation $w_j = c_j$ lies in the g-core for every θ, and g is the least universal blocking-immunity factor. (iii) There are instances whose transferable-utility core is empty while the $\theta = 0$ core is nonempty—a separation driven by overlap, like the integrality gap.*

This is one of three appearances of a single phenomenon: the no-numeraire constraint *removes deviations* (VCG payments become infeasible; batch deviations are dominated at $g = 1$; componentwise blocking is weak). Whether the $\theta = 0$ core is always nonempty is open: Scarf's route needs balancedness, broken by the same overlap geometry.

Complexity. Fair winner determination is NP-hard (from 3-dimensional matching) but fixed-parameter tractable in the number P of directed asset pairs: UDP forces one bid per pair, so $2^{O(P)} \cdot \mathrm{poly}$ suffices. The deployment cap on pairs per batch thus makes the problem tractable in exactly the deployed regime.

5 The Price of Strategyproofness

The impossibility of fairness-compatible strategyproof mechanisms with *endogenous* floors (VCG is infeasible without transfers; an online incompatibility, Section 6) motivates an *exogenous-benchmark* regime: floors come from a public, g-calibrated historical benchmark, and solvers are single- or multi-minded with private values passing the filter. We measure the price of strategyproofness PoSP, the welfare ratio between FAIR and the best universally truthful fair mechanism.

The core of the regime is already nontrivial for one strategic solver:

Theorem 4 (Core characterization). *Over the screening core R_g (one strategic solver, public floors), the price of strategyproofness is $\Theta(1 + \log g)$: every randomized DSIC mechanism pays at least $\frac{g \ln g}{2(g-1)} \geq \frac{\ln g}{2}$, a geometric posted-price mechanism achieves $e(\lceil \ln g \rceil + 1)$, and the best deterministic mechanism is exactly $\sqrt{g}$.*

The lower bound is a model-native technique: a one-sided envelope inequality (downward incentive compatibility alone forces monotone utilities, hence $u' \geq p$ a.e.) integrated against a V^{-2} kernel. Two mechanisms lift this to the full game.

Theorem 5 (Two routes to $\Theta(1 + \log g)$). *(i) For arbitrarily many single-minded solvers with public floors, an eligibility-posted packing mechanism is universally truthful and $\Theta(1 + \log g)$-competitive. (ii) For one multi-minded solver of arbitrary structure, a coupled-menu mechanism—post all options at prices $C_S\, t$ for a single random multiplier $t = e^k$ and let the solver demand any profit-maximizing disjoint collection—is universally truthful and $\Theta(1 + \log g)$-competitive. Independent per-option pricing instead pays $\Omega(g/\log^2 g)$.*

The multi-minded bound rests on a *profit-curve* technique: the solver's profit $p(t)$ over disjoint collections is convex and piecewise linear, delivered mass is $t\,(-p'(t))$, and a geometric grid telescopes $p(1)$ against expected welfare. The remaining cell is both multi-agent and multi-minded:

Theorem 6 (Multi-agent multi-minded). *Random-priority coupled menus are universally truthful with $\mathbb{E}[W] \geq \mathrm{FAIR}/\big(N(e-1)(\lceil \ln g \rceil + 1) + 1\big)$, i.e. $\Theta_N(1 + \log g)$ for a constant number N of strategic solvers. Conversely every serial-demand mechanism—a report-independent ex-ante service order, arbitrary per-identity multiplier distributions, each agent demanding a profit-maximizing collection—pays $\Omega(\min(N, g))$, even with geometry-aware ordering. So any N-independent* $\mathrm{polylog}(g)$ *mechanism must break the serial-demand format.*

The barrier is proved by a *symmetrized swarm*: all identities share the same public geometry $\{\{o_i\}, G\}$, differing only in private values, so no report-independent order can tell the batcher from the blockers; a telescoping identity $\sum_i (1 - q_i) \prod_{s \prec i} q_s = 1 - \prod_s q_s \leq 1$ then places the batcher behind an expected blocker.

6 Online and Strategic Prices

In a lifetime-scoped benchmark model (orders have lifetimes; floors are benchmarked within them), settle-at-deadline gives:

Proposition 3 (Deterministic online ratio). *The deterministic online competitive ratio of fair settlement is exactly g.*

Theorem 7 (Two online prices). *(i) Against realized lifetime floors with almost-sure fairness, randomization does not help: the competitive ratio is exactly g (an exposed batch settled with positive probability is undercut by a later standalone bid). (ii) On static single-contention instances (public floors, fleeting pairwise-conflicting offers), the deterministic ratio is $\Theta(\sqrt{g})$ and the randomized ratio is $\Theta(\log g)$, the latter by one-way search over the spread on the stopped welfare $q = V + C - C_S \in [C, gC]$. Ratio-threshold greedy mechanisms pay $\Omega(g)$: the swarm of Theorem 6 in online clothing.*

The geometric-level hard family here is the same scale-free family that governs R_g offline: one family prices both the informational and the incentive cost of the spread g.

The strategic face. For the first-price (pay-as-bid) game, measured in *produced* welfare PW (the delivered objective is unboundedly manipulable), the individual-only restriction is smooth.

Theorem 8 (Individual competition is smooth). *In the per-order-decomposable individual-only game, every coarse correlated equilibrium has $\mathbb{E}[\mathrm{PW}] \geq (1 - \frac{1}{e})\mathrm{OPT}_{\mathrm{ind}}$, so its price of anarchy is at most $e/(e-1)$. The model-native ingredient is that floors make standing bids coincide with delivered prices, $c_j(b) = p_j(b)$, so the Syrgkanis–Tardos penalty lands under the original profile. Against the unrestricted optimum this gives $(1 - 1/e)\mathrm{OPT}/g$. A weak pure equilibrium realizes the lower bound g; the full-game price of anarchy is conjectured $\Theta(g)$, obstructed by latent batches that execute only under the deviation.*

Pricing manipulation. Finally we price manipulation of the benchmark itself. On one directed pair with honest quantile benchmark b^*, an attacker of volume v injects contamination mass a at convex cost $K(a) = \kappa_1 a + \frac{\kappa_2}{2} a^2$, depressing the linearized benchmark by βa with $\beta = 1/(mf(b^*))$; the published benchmark falls by at most a factor $1 - r$ per round (downward rate limit, instant recovery).

Theorem 9 (The contamination game). *A marginal injection cost $\kappa_1 \geq v\beta$ deters contamination entirely; otherwise the optimal constant attack is $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$ with steady relative bias $\mu = \beta a^*/b^*$. The pointwise bound $b_t \geq b^* - \beta a_t$ makes bursts no better than the constant attack, and the rate limit is a pure ramp tax: it leaves the steady state unchanged but taxes restarts through the finite ramp. Steady contamination feeds every floor-calibrated guarantee through $g \mapsto g/(1-\mu)$.*

Detection only tempers intensity; deterrence is the cost floor's job. The slogan: fees deter, detection tempers, rate limits tax ramping but do not remove persistent bias.

7 Conclusion

A single invariant—the complementarity factor g, with liquidity θ—governs the worst-case fairness, equilibrium, complexity, incentive, online, and manipulation prices of the no-numeraire fair combinatorial auction. Two phenomena recur: the no-numeraire constraint removes deviations (aiding equilibrium and the core but disabling VCG), and one geometric family prices both informational and incentive costs of the spread. Two obstructions are named rather than hidden—latent batches in the full-game price of anarchy, and the serial-demand barrier in the multi-agent multi-minded cell—marking the natural next targets.

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A Full Proofs for Section 3

Proof (Proof of Theorem 1). Upper bound. Fix an unconstrained optimum and process its winning bids independently; bids cover disjoint order sets, so the repaired welfares add. Take a winning bid with floor mass $C_S = \sum_{j\in S} c_j > 0$ (a bid with $C_S = 0$ has all floors zero and is already fair), value $V = \sum_{j\in S} v_j$, gain $\gamma = V/C_S \in [1, g]$, and deficit $D = \sum_{j\in S}(c_j - v_j)^+$. Two fair uses:

Repair. Move value from over-served legs to starved ones. The total surplus available is $s = V - (C_S - D) = V - C_S + D$, and clearing the deficit costs D/θ units of surplus (haircut $\lambda = 1-\theta$). This is feasible iff $V - C_S \geq D\frac{1-\theta}{\theta}$, and then the delivered welfare is $f = V - D\frac{1-\theta}{\theta}$. Since $D \leq C_S$ and $V - C_S = (\gamma-1)C_S$, feasibility holds whenever $\gamma - 1 \geq \frac{1-\theta}{\theta}$, i.e. $\gamma \geq 1/\theta$; then $f \geq V - C_S\frac{1-\theta}{\theta} = V\big(1 - \frac{\gamma-1+\cdots}{}\big)$, and a direct computation gives $f \geq V/\min(g, 1/\theta)$ in the regime $\gamma \geq \min(g, 1/\theta)$.

Drop. Settle every covered order at its floor: $f = C_S = V/\gamma$. For $\gamma \leq \min(g, 1/\theta)$ this is $\geq V/\min(g, 1/\theta)$.

Either way each bid retains a $1/\min(g, 1/\theta)$ fraction of its value; summing, $\text{FAIR} \geq \text{OPT}/\min(g, 1/\theta)$.

Lower bound. Let $\gamma^* = \min(g, 1/\theta)$. Two orders on distinct pairs, $c_1 = 0$, $c_2 = c$; one batched bid with $v_2 = 0$, $v_1 = \gamma^* c$, so $V = \gamma^* c$, $C_S = c$, and the batching solver's standalone values are $0, c$, so Definition 1 holds as $\gamma^* \leq g$. Repairing order 2 to its floor c requires withdrawing c/θ from v_1; if $\gamma^* = 1/\theta$ this exhausts v_1 exactly (welfare c), and if $\gamma^* = g < 1/\theta$ the batch is irreparable (welfare c from the individual allocation). In both cases $\text{FAIR} = c$ and $\text{OPT} = \gamma^* c$, so $\text{OPT}/\text{FAIR} = \gamma^*$. Independence of the number of orders and of prices is immediate from the construction. □

Proof (Proof of Proposition 1). The two-order family of Theorem 1 already has $d = 2$ and attains $\min(g, 1/\theta)$, and the upper bound never uses d; for $d = 1$ only singletons exist, so $\text{OPT} = \text{FAIR} = C$. □

Proof (Proof of Theorem 2). Upper bound. As above, coverage caps the deficit by $D \leq (1-\alpha)C_S$. Repair is feasible when $\gamma - 1 \geq (1-\alpha)\frac{1-\theta}{\theta} = \mathit{\Gamma} - 1$, and then $f = V - \frac{D(1-\theta)}{\theta} \geq V\big(1 - \frac{(1-\alpha)(1-\theta)}{\theta\gamma}\big) \geq V/\mathit{\Gamma}$ for $\gamma \geq \mathit{\Gamma}$ (monotone in γ, equality at $\gamma = \mathit{\Gamma}$); dropping gives $V/\gamma \geq V/\mathit{\Gamma}$ for $\gamma \leq \mathit{\Gamma}$ and V/g for $\gamma \leq g$. Hence $\text{FAIR} \geq \text{OPT}/\min(g, \mathit{\Gamma})$. *Lower bound.* Two orders, $c_1 = 0$, $c_2 = c$; batched bid with $v_2 = \alpha c$ (coverage saturated) and $v_1 = (\gamma^* - \alpha)c$, $\gamma^* = \min(g, \mathit{\Gamma})$. Repairing order 2 needs $(1-\alpha)c/\theta$ from v_1; if $\gamma^* = \mathit{\Gamma}$ then $v_1 = (\mathit{\Gamma} - \alpha)c = \frac{(1-\alpha)c}{\theta}$ exactly, and if $\gamma^* = g < \mathit{\Gamma}$ the batch is irreparable. Either way $\text{FAIR} = c$, $\text{OPT} = \gamma^* c$. □

B Full Proofs for Section 4

Proof (Proof of Lemma 1). $\sum_{j\in S}(v_j - w_j) = V_i(S) - \sum_{j\in S} w_j \leq \sum_{j\in S} V_i(\{j\}) - \sum_{j\in S} w_j \leq \sum_{j\in S}(V_i(\{j\}) - w_j)^+$, the middle step by $g = 1$. Each positive-profit singleton has $V_i(\{j\}) > w_j$, hence is executable; a non-executable batch is not a deviation. The obligation w_j is the same whether j is served in the batch or individually, which is the only property of UDP used. □

Proof (Proof of Theorem 3). (a), $g = 1$. Set $w_j = c_j$ and assign each order j to a solver attaining $V_i(\{j\}) = c_j$. The winner earns zero margin per order; any losing singleton has $V_i(\{j\}) - c_j \leq 0$; and Lemma 1 kills batch deviations (no positive-profit singletons). With $g = 1$, any allocation produces at most $\sum_{(i,S)} V_i(S) \leq \sum_j c_j = C$, attained here, so produced value is $C = \mathrm{OPT}$, all delivered. The point $w_j = c_j$ is the best-bid floor, so the equilibrium is fair (the trader-optimal core point of the underlying assignment market [9], to which Lemma 1 reduces the economy).

(b), $g > 1$. We argue natively; non-existence with transfers does not transfer, as feasibility only removes deviations. Fix $\delta \in (0, 2(g-1)]$; orders $1, 2, 3$ on distinct pairs, a rival with standalone value 1 on each ($c_j = 1$), and three solvers each holding one batched bid on a pair $\{j, j'\}$ with vector $(1 + \frac{\delta}{2}, 1 + \frac{\delta}{2})$ (value $2 + \delta \leq 2g$) and standalone values 1. The efficient allocation runs one batch, say $\{1, 2\}$, plus order 3 individually (produced $3 + \delta$). Suppose rates w support it. Order 3 is served by a standalone solver of value 1, so $w_3 \leq 1$; orders $1, 2$ are served by the batch, so $w_1, w_2 \leq 1 + \frac{\delta}{2}$. The batch $\{2, 3\}$ is then executable, and equilibrium requires it unprofitable against its solver's individual alternatives: $(2 + \delta) - w_2 - w_3 \leq (1 - w_2)^+ + (1 - w_3)^+$. If $w_2 \leq 1$ the right side is $(1 - w_2) + (1 - w_3)$, giving $\delta \leq 0$; if $w_2 > 1$ it is $1 - w_3$, giving $w_2 \geq 1 + \delta > 1 + \frac{\delta}{2}$, contradicting feasibility. No supporting rates exist. □

Proof (Proof of Proposition 2). (i) If $w_j < c_j$, the standalone bid attaining c_j blocks j at factor 1; if $w_j \geq c_j$ for all j, no singleton strictly improves. (ii) Blocking $w_j = c_j$ at factor g needs delivery $> g\, c_j$ to every $j \in S$, totalling $> g\, C_S$, while any feasible delivery from a bid totals $\leq V_i(S) \leq g\, C_S$ (haircuts only lose value; if all floors in S are zero then $V_i(S) = 0$ and strict blocking is impossible). Tightness: a batch with vector $(g'c, g'c)$ over floors (c, c), $g' \in (\kappa, g]$, blocks at any $\kappa < g$. (iii) Three-cycle floors $(1, 1, 1)$, pair vectors $(1 + \frac{\delta}{2}, 1 + \frac{\delta}{2})$: summing the three transferable core constraints $x_j + x_{j'} \geq 2 + \delta$ gives $\sum x_j \geq 3 + \frac{3\delta}{2} > 3 + \delta = \mathrm{OPT}$, so the transferable core is empty; the allocation (batch $\{1, 2\}$, order 3 individually) is $\theta = 0$-unblocked since every other batch caps a covered order at a value already attained. □

Complexity (sketch). NP-hardness is a reduction from 3-dimensional matching: orders are ground elements, batched bids are triples of value 1, and a fair welfare-n allocation exists iff a perfect matching does. For fixed-parameter tractability, UDP forces all orders on a directed pair into one winning bid, so a solution is a choice of one bid per pair; enumerating over $2^{O(P)}$ pair-assignments and solving each in polynomial time gives $2^{O(P)} \cdot \mathrm{poly}$.

C Full Proofs for Section 5

Proof (Proof of Theorem 4 (sketch)). Lower bound. Restrict to the screening core R_g: a single strategic solver with a batch type and public fallback floors. Downward incentive compatibility forces the interim utility u to be nondecreasing, hence $u'(V) \geq p(V)$ almost everywhere (a one-sided envelope inequality—two-sided Myerson fails because binding bids misreport only downward). Integrating $u' \geq p$ against the kernel V^{-2} and applying Fubini yields expected welfare loss $\geq \frac{g \ln g}{2(g-1)} \geq \frac{\ln g}{2}$ for every randomized DSIC mechanism. *Upper bounds.* A geometric posted-price ladder achieves $e(\lceil \ln g \rceil + 1)$; the optimal deterministic mechanism is exactly $\sqrt{g}$ by a balanced two-threshold argument. Since R_g embeds in the full model, the lower bound holds there too. ◻

Proof (Proof of Theorem 5 (sketch)). (i) The eligibility-posted packing mechanism posts, from public data, an eligibility threshold and packing weight $C_S e^k$ per option for a random k; it is universally truthful (no reports enter the posted quantities) and a $\max(X/a, Y) \geq (X+Y)/(a+1)$ combination gives $\mathbb{E}[W] \geq \mathrm{FAIR}/(e(\lceil \ln g \rceil + 1) + 1)$. (ii) Post every option at price $C_S\, t$, $t = e^k$, $k \sim \mathrm{Unif}\{0, \ldots, K\}$; the solver demands a profit-maximizing disjoint collection. Truthfulness is in the posted-demand sense. Let $p(t) = \max_{\mathcal{C}} \sum_{S \in \mathcal{C}} (V_S - C_S t)$ over disjoint collections: it is convex, piecewise linear, nonincreasing, and 0 for $t \geq g$; at differentiability $-p'(t) = \sum_{S \in \mathcal{C}^*(t)} C_S$, so delivered mass is $D(t) = t(-p'(t))$. On $[e^k, e^{k+1}]$, $\int -p' \leq (e-1)D(e^k)$; telescoping, $p(1) \leq (e-1)\sum_k D(e^k)$, whence $\mathbb{E}[D] \geq p(1)/((e-1)(K+1))$, and combining with $W \geq C$ gives $\Theta(1 + \log g)$. For the separation, m options sharing one order under independent geometric prices let a multi-minded solver cherry-pick the cheapest, losing $\Omega(g/\log^2 g)$. ◻

Proof (Proof of Theorem 6 (sketch)). Upper bound. In random-priority coupled menus, condition on a solver being served first (probability $1/N$): it faces the full coupled menu and the profit-curve telescope of Theorem 5 applies, while later positions contribute nonnegatively; since each solver's optimal batches form a disjoint collection, $\sum_i p_i(1) \geq \sum_b V_b - C_{B^*}$, and the bound follows. Truthfulness holds because a solver's menu depends only on predecessors' demands. *Lower bound.* Symmetrized swarm: N identities, each with public family $\{\{o_i\}, G\}$ (G all orders), differing only privately; the batcher (placed at an adversarial identity after seeing the mechanism) has $V(G) = \rho C_G$, while blockers have a profitable ρ-singleton. Choose $\rho \in (\max\{1, g/2\}, g]$ off all multiplier atoms. A blocker is silent iff $t > \rho$; the telescoping identity $\sum_i (1 - q_i) \prod_{s \prec i} q_s = 1 - \prod q_s \leq 1$ yields an identity i^* with $(1 - q_{i^*})\Pr[\text{predecessors silent}] \leq 1/N$, and placing the batcher there caps its expected delivery at $2\rho L/N$, giving ratio $\Omega(\min(N, g))$. ◻

D Full Proofs for Section 6

Proof (Proof of Proposition 3). Settle-at-deadline is fair and achieves C on every instance, while $\mathrm{FAIR} \leq \mathrm{OPT} \leq gC$, so the ratio is at most g; a staggered-deadline instance forcing individual-only settlement shows it is at least g. ◻

Proof (Proof of Theorem 7). (i) If an almost-surely fair mechanism settles an exposed batch (a covered order's deadline in the future) with positive probability on some prefix, fix that prefix and append a higher standalone bid on a later-deadline covered order: on the appended instance it has, with positive probability, settled below a realized floor—not a.s. fair. Hence exposed batches are never settled, and on the staggered hard family the mechanism is individual-only a.s., with welfare $\leq C$ against FAIR $= gC$. (ii) Stopped welfare $q = V + C - C_S$ satisfies $q \in [C, gC]$ ($V \geq C_S$ and $V \leq gC_S$, $C_S \leq C$). *Deterministic* $\Theta(\sqrt{g})$: accept the first $q \geq \sqrt{g}\,C$; if $q^* \geq \sqrt{g}\,C$ the ratio is $q^*/q \leq \sqrt{g}$, else $W = C$ and ratio $< \sqrt{g}$; a two-step adversary matches. *Randomized* $O(\log g)$: with $k \sim \mathrm{Unif}\{0, \ldots, K\}$, accept the first $q \geq e^k C$. With probability $1/(K+1)$, $e^k C \leq q^* < e^{k+1}C$, and then the mechanism accepts some offer with $q \geq e^k C > q^*/e$, so $\mathbb{E}[W] \geq q^*/(e(K+1))$. The matching $\Omega(\log g)$ is a Yao bound: a persistent order with fleeting geometric batch offers $x_i = e^i c$, $i \leq \lfloor \ln g \rfloor$, shown for the first M with $\Pr[M \geq i] = e^{-i}$, caps every deterministic threshold at $2c$ while $\mathbb{E}[\mathrm{FAIR}] = \Omega(c \log g)$. Greedy: a ratio-$g$ blocker arriving before a ratio-g batch is accepted (killing the batch) for any threshold $T \leq g$ and rejected with the batch for $T > g$, ratio $\Omega(g)$. □

Proof (Proof of Theorem 8 (sketch)). In the per-order-decomposable individual game, the maximal standing bid on j both wins and delivers itself, so $c_j(b) = p_j(b)$. Let $i^*(j)$ deviate to W_j with density $1/(v_j^{\max} - w)$ on $[0, (1 - 1/e)v_j^{\max}]$ (Syrgkanis–Tardos). Then the expected utility from j is $((1 - \frac{1}{e})v_j^{\max} - \beta_j)^+ \geq (1 - \frac{1}{e})v_j^{\max} - p_j(b)$ with $\beta_j \leq c_j(b) = p_j(b)$, the penalty landing under b. Summing and using $\sum_i u_i(b) = \mathrm{PW}(b) - \sum_j p_j(b)$ in the CCE inequality cancels the price terms, giving $\mathbb{E}[\mathrm{PW}] \geq (1 - 1/e)\mathrm{OPT}_{\mathrm{ind}}$; against the unrestricted optimum, $\mathrm{OPT} \leq g\,\mathrm{OPT}_{\mathrm{ind}}$. The pure-NE lower bound is Theorem 1's instance with a filtered $(2gv, 0)$ batch. The full-game obstruction is a latent batch present in b_{-i} that executes under the deviation but not under b, breaking every charging scheme. □

Proof (Proof of Theorem 9 (sketch)). The constant-policy per-round profit is $\pi(a) = v\beta a - \kappa_1 a - \frac{\kappa_2}{2}a^2$ (the rate limit does not bind in steady state). With $\kappa_2 > 0$: if $v\beta \leq \kappa_1$ then $\pi'(0) \leq 0$ and π is concave, so $a^* = 0$; else $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$. The dynamics give $b_t \geq b^* - \beta a_t$ pointwise, so every horizon-T payoff is $\leq \sum_t [v\beta a_t - K(a_t)] \leq T\pi(a^*)$; the constant attack attains this after the ramp. Reaching bias μ needs $\geq \ln \frac{1}{1-\mu} / \ln \frac{1}{1-r}$ rounds, during which the benefit is below steady state; with instant recovery any pause restarts the ramp, so the exact net loss per restart is $\sum_{t=1}^{T_\mu} [\pi(a^*) - \pi(\Delta_t/\beta)] \approx \pi(a^*)\mu/(3r)$. Floors scale by $1 - \mu$, so a guarantee in the public-floor spread G becomes $G/(1-\mu)$; the coverage law calibrates jointly, $(g, \alpha) \mapsto (g/(1-\mu), \min\{1, \alpha/(1-\mu)\})$. □